

Thm for a linear map $f: V \rightarrow W$ between f.dim'l V spaces

we have $\dim(\ker f) + \dim(\operatorname{im} f) = \dim V$

$$\begin{array}{ccc} n \swarrow & \downarrow & \swarrow r \\ \{v \mid f(v)=0\} & & f(V) \\ & \downarrow & \downarrow d \\ & & d \end{array}$$

Pf [Example: $\mathbb{R}^3 \rightarrow \mathbb{R}^4$ $\ker = \left\{ \begin{bmatrix} 0 \\ 0 \\ c \end{bmatrix} \mid c \in \mathbb{R} \right\} \rightarrow \text{has dim 1}$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \rightarrow \begin{bmatrix} a \\ b \\ 0 \\ 0 \end{bmatrix}$$

Im has dim 2.

Pf in general situation will be reduced to such simple situation by choosing a good basis!

Pf. Let $\dim \ker f = n$. Choose basis $\underbrace{k_1, \dots, k_n}_{\substack{\uparrow \\ \text{to } v_1, \dots, v_{d-n}}}$
Let $\dim V = d$. Add $v_1, \dots, v_{d-n}$ to $\uparrow$ to get a
basis $\{k_1, \dots, k_n, v_1, \dots, v_{d-n}\}$ of V

Want: $d-n =$
 $\dim$ of $f(V)$

We will be done by finding a basis of $f(V)$ of cardinality $d-n$.

Idea: Try to show that $f(v_1), \dots, f(v_{d-n})$ form a basis of $f(V)$.

(Proving this will finish the proof!)

Why do they span $f(V)$? Given $v \in V$, write $v = a_1 k_1 + \dots + a_n k_n + b_1 v_1 + \dots + b_{d-n} v_{d-n}$

Why are $f(v_1), \dots, f(v_{d-n})$ indep?

note: $f(v) = f(\dots) = f(\sum a_i k_i) + \sum_{i=1}^{d-n} b_i f(v_i) \checkmark$

$$\text{Let } c_1 f(v_1) + \dots + c_{d-n} f(v_{d-n}) = 0.$$

$$\therefore f(c_1 v_1 + \dots + c_{d-n} v_{d-n}) = 0$$

$$\therefore c_1 v_1 + \dots + c_{d-n} v_{d-n} \in \ker(f)$$

$$\therefore c_1 v_1 + \dots + c_{d-n} v_{d-n} = p_1 k_1 + \dots + p_n k_n = 0 \text{ for some } p_i, \therefore p_n$$

$$\therefore c_1, \dots, c_{d-n} \text{ (and } p_1, \dots, p_n) \text{ are all } 0. \text{ Done!}$$

One line summary of proof: "Extend a basis of kernel of f ."

Ex. Can you give a proof that uses a given basis of $\text{Im}(f)$?

Question asked
in class.
Work out the
details!

→ Ex. We already
proved rank-nullity
thm for $f: \mathbb{R}^c \rightarrow \mathbb{R}^r$
 $x \mapsto Ax$

(by row reduction
and interpretation of
 $\dim(\ker f) = \# \text{ free var}$
 $\dim(\text{im } f) = \# \text{ pivots}$)

Can you reduce general
case to that situation?

Yes! Use $V \xrightarrow{f} W$
 $\downarrow \quad \downarrow$
isomorphisms $\mathbb{R}^d \dashrightarrow \mathbb{R}^m$

Calculating with coordinates

Recall. Given ordered basis $B = (v_1, \dots, v_n)$ of V ,

we get linear map $\mathbb{R}^n \rightarrow V$

which is an isom.

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \mapsto \sum c_i v_i$$

Its inverse is the coordinate map

$$[v]_B \longleftarrow v$$

Answer: i th column of this matrix is $[f(v_i)]_C$

← Question: which matrix??

Take V with basis $B = \{v_1, \dots, v_n\}$

W with basis $C = \{w_1, \dots, w_k\}$

$$v \in V \xrightarrow{f} W \ni f(v)$$

$$\begin{array}{ccccc} \downarrow \alpha & & \downarrow & & \downarrow \beta \\ \mathbb{R}^n & \xrightarrow{\beta f \alpha^{-1}} & \mathbb{R}^k & & \\ \downarrow & \text{??} & \downarrow & & \\ [v]_B & \xrightarrow{\quad} & [f(v)]_C & & \end{array}$$

← $\beta f \alpha^{-1}$ must be mult. by a matrix!